

Monty Hall Shock Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	monty-hall-shock

Big idea: Formal probability can beat intuition.

Overview

Students feel the conflict between intuition and formal probability.

Student Prompt

Inspect Monty Hall Shock Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Monty Hall Shock Stimulus A	Student-facing stimulus 1: Three paper doors, one prize card, and repeated rounds with switch/stay tracking. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Monty Hall Shock Stimulus B	Student-facing stimulus 2: A class results table showing wins after switching versus staying. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Monty Hall Shock Reveal / Twist	Student-facing stimulus 3: A probability tree showing that the initial choice has probability $\frac{1}{3}$ and the two unchosen doors together have probability $\frac{2}{3}$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Formal probability can beat intuition.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Monty Hall Shock Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Monty Hall Shock Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Why does mathematical reasoning often feel wrong at first?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how formal probability can beat intuition.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: Three paper doors, one prize card, and repeated rounds with switch/stay tracking.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- Three paper doors, one prize card, and repeated rounds with switch/stay tracking.
- A class results table showing wins after switching versus staying.
- A probability tree showing that the initial choice has probability $1/3$ and the two unchosen doors together have probability $2/3$.

Suggested Timing

Stage	Teacher move	Watch for
1	Play the Monty Hall game physically. A student chooses a door.	Assumptions, confidence shifts, evidence claims, and language choices.
2	The host opens an empty door.	Assumptions, confidence shifts, evidence claims, and language choices.
3	The student chooses whether to stay or switch.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Repeat many times and compare win rates for stay and switch.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Proof versus Pattern
- Axiom Switch Universe
- Calculator Trust Test